

Tutorial: Inequality and Poverty in Canada

ECON 204 – Contemporary Economic Issues

Winter 2026

Question 1. Income Distribution and Inequality

Consider the following **annual before-tax household incomes** (in Canadian dollars) for **10 households**, ordered from lowest to highest income:

Household	Income (\$)
1	18,000
2	22,000
3	26,000
4	30,000
5	35,000
6	42,000
7	50,000
8	65,000
9	95,000
10	180,000

(a) Mean and Median Income

- Calculate the **mean (average)** household income.
- Calculate the **median** household income.

Explain which measure better reflects the “typical” household income in Canada and why.

(b) Income Shares

- Calculate total income across all households.
- Calculate the income share of:
 - the **bottom 40%** (households 1–4),
 - the **top 10%** (household 10).

Use the formula:

$$\text{Income Share} = \frac{\text{Group Income}}{\text{Total Income}} \times 100$$

Interpret what these shares imply about income concentration.

Question 2. Simple Inequality Ratios

(a) Top-to-Bottom Ratio

Calculate the ratio:

$$\text{Top-to-Bottom Ratio} = \frac{\text{Income of richest household}}{\text{Income of poorest household}}$$

Explain what this ratio indicates about inequality.

(b) Policy Change

Suppose a progressive tax-and-transfer policy reduces the richest household's income to \$150,000 and raises the poorest household's income to \$25,000.

- Recalculate the top-to-bottom ratio.
- State whether inequality has increased or decreased.

Question 3. Poverty Measurement in Canada

Assume the following annual poverty thresholds for a **single adult living in a large Canadian city**:

- Low-Income Cut-Off (LICO): \$26,000
- Market Basket Measure (MBM): \$28,000

(a) Poverty Status

Using the income data from Question 1:

- How many households fall below the LICO?
- How many households fall below the MBM?

(b) Poverty Rates

Calculate the poverty rate under each measure:

$$\text{Poverty Rate} = \frac{\text{Number of households below the threshold}}{\text{Total households}} \times 100$$

Explain why the poverty rate differs across measures.

Question 4. Depth of Poverty

For households below the MBM threshold:

(a) Poverty Gap

For each poor household, calculate:

$$\text{Poverty Gap}_i = \text{MBM Threshold} - \text{Household Income}_i$$

(b) Average Poverty Gap

Compute the average poverty gap among households below the MBM.

Explain why the poverty gap is an important policy-relevant measure.

Question 5. Inflation and Poverty

Suppose inflation increases the MBM threshold by **8%**, while nominal household incomes remain unchanged.

(a) New Poverty Line

Calculate the new MBM threshold.

(b) Poverty Impact

- How many households now fall below the new MBM threshold?
- Explain how inflation can increase poverty even if nominal incomes do not change.